



alient12

AliEnt

Ali Entezari

COMPUTER SKILLS

Programming/Scripting

- Python
 - Tensorflow
 - Pytorch
 - Sklearn
 - PyQt
 - OpenCV
 - Matplotlib
 - Selenium
- C/C++
 - STL
 - MemoryManagement
 - LibPatch
 - Capstone
 - OpenGL
 - Vulkan
- JavaScript
- TypeScript
- ReactJS
- HTML/CSS
- Golang
 - JWT
 - SQL & NoSQL
 - Microservices
 - Websocket
- VHDL
- MATLAB
 - Basic Codes
 - App Designer
 - Simulink
- Assembly
- L^AT_EX

Tools

- Tracing
 - LTTng
 - BPFtrace
 - GDB
 - uftrace
 - strace
- MongoDB
- Linux
- ROS
- SolidWorks
- MSC ADAMS
- Simscape
- STM32 Cube MX
- Docker
- Git
- CI/CD
- Proteus
- Vivado

Hardwares

- STM32 Microcontrollers
- Nvidia Jetson
- Raspberry Pi
- Arduino
- FPGA
- AVR

EDUCATION

Research-based Master's



Polytechnique Montréal

2025–present
Montréal-Canada

- Computer and Software Engineering
 - Research Field: Software Testing and Tracing - [REDACTED]
 - GPA: 3.86/4

Bachelor of Science



Amirkabir University of Technology (Tehran Polytechnic)

2019–2024

Tehran-Iran

- Electrical and Control Engineering
 - Major: Control Engineering
 - **Thesis:** Development of a Delta Parallel Robot for Computer Vision-Based Fruit Inspection and Sorting - [REDACTED]
 - Supervisor: [REDACTED]
 - GPA: 3.45/4 - 16.88/20 - JASSO: 2.44
 - GPA (last 2 years): 3.47/4 - 16.87/20 - JASSO: 2.46

Diploma of Mathematics and Physics



Danesh School

2013–2019

Tehran-Iran

WORK AND INTERNSHIP EXPERIENCE

Research Assistant - Nagoya University

Summer 2025

- Projects:
 - Development of a Computer Vision System for Automatic Customer Reaction Detection in Teleoperated Social Robots

Nagoya - Japan

Developer - BEC Afzar

2021 – 2023 2 yr

- Projects:
 - Machine Learning Based Controller

Tehran - Iran

- UPS Monitoring system over wifi
-  AI Developer - *iTherMAl*
 - Computer Vision projects (Object detection, Image Segmentation and ...)
 - Nvidia Jetson Optimization & CI/CD

Jul 2021 – Nov 2022 1 yr 6 mos

Brussels - Belgium

SELECTED PROJECTS

- Conditional Dynamic Control Flow Tracer
 - A Tool that helps you to debug your program by recording CFG dynamically based on certain conditions.
 - [Github link](#)
- Digital Twin of Kinova Gen3 Robot Arm
 - A digital twin for analyzing trajectory deviation of a robot arm with 3d digital shadow of the model and robot simulation for singularity detection and emergency stop
 - [Github link](#)
- SwabAI (open source wide open mouth detector)
 - The mouth coordinates are found in a more certified way using two independent algorithms (YOLOv5n and Dlib). Therefore, the algorithm can be used in more sensitive applications. Accelerated using TensorRT for realtime usage.
 - [Github link](#)
- Smoke and Fire Detection using Computer Vision (Classic + ML)
 - Smoke and fire detection for indoor industrial environments. Optimised for Nvidia Jetson boards.
 - [iTherMAl website](#)
- Heartbeat Detection using Image processing
 - The algorithm finds ROI (forehead and cheeks) of face by using Dlib and then calculates the average of Red-RGB of ROI pixels. Afterwards deep learning is used on Red-RGB signals.
 - [iTherMAl website](#)
- Solving Water Sort Puzzle
 - Using python OpenCV to detect number of bottles and order of colors in each bottle and prepare input for solver program.
 - Solving game using DFS in C++
 - [Github link](#)
- DC Motor Control with GUI Control Panel
 - DC motor PID controller on STM32 using C
 - Connecting python GUI to STM32 using USB
 - Realtime plotting and transfer function estimator
 - [Github link](#)

COURSES

- | | |
|--|---|
| ○  Intelligent DevOps of Large-Scale Software Systems | ○  Release Eng.App. Mining Software Repositories |
| ○  Digital Twin Engineering | ○  Software Verification & Testing |
| ○  Advanced Programming | ○  Web Programming |
| ○  Introduction to Computational Intelligence | ○  Intermediate Machine Learning Course |
| ○  Introduction on Computational Intelligent Lab | ○  Computer Architecture & Microprocessors |

HONORS

- | | |
|---|------|
| ○ Passed First level of Physics Olympiad | 2018 |
| ○ Passed First level of Astronomics Olympiad | 2018 |

LANGUAGE SKILLS

- Persian Native
- English Academic IELTS - Band Score: **7.0** (Listening: 6.5 Reading: 7.0 Writing: 7.0 Speaking: 6.5)
- French CEFR Level A2
- Japanese CEFR Level A1

📌 References, Further information, and Proofs are available upon Request